

International Islamic University Chittagong
Department of Computer Science and Engineering
Report on Final Year Project/Thesis as Complex Engineering Problem

The following report will justify that the work done in your project/thesis is a solution to a complex engineering problem. To do this, firstly, specify which attributes of complex engineering problem (P) are present in the problem you have formulated and solved. Secondly, specify which complex engineering activities (A) are involved when solving the problem, particularly when you required to communicate for solving the problem. Provide explanation for each P and A involved. Finally, give an overall comment justifying that you have solved a complex engineering problem in your project/thesis.

Title of the Project/Thesis:	CacheFlow: Memory-Aware Expert Scheduling for Efficient Mixture-of-Experts Models	
Members of the Group	Name	ID
	Mir. Md. Tarhimul Quader	C221017
	Md. Iftaker Ahamed Sayem	C221020
	Turja Dutta	C221026
Name of the Supervisor	Jamil As-Ad	

Which attributes of the Complex Engineering Problems (P) are present?

Attribute of Complex Engineering Problem	Explanation (Justify that the attribute was present and you have addressed it sufficiently)
P1: Depth of knowledge required	The project required a wide range of skills in a variety of advanced fields: deep learning architectures and transformer-based large language models and Mixture-of-Experts systems and Apple-low memory management and methods of optimizing inference. It was necessary to know how experts are directed and how key-value caches, the impact of quantization and runtime scheduling to address the memory issues of large MoE models. This represents a significant amount of interdisciplinary experience that is way beyond the common undergraduate courses, which would prove invaluable in addressing the memory-inefficiency problem.
P2: Range of conflicting requirements	The test task was to minimize the memory consumption of GPUs and maintain inference constant and quick. Eradicating experts of memory may reduce the amount of memory that is used, however, would raise the latency and would render the execution vulnerable. Retaining all specialists would swamp the GPU. CacheFlow eases this tension with an addition of a priority-based scheduler, which balances the memory constraints of the performance requirements.

P3: Depth of analysis required (no obvious solution)	No existing solution was developed to execute big MoE models on hardware with a small memory. The issue involved submerging into the understanding of accessing experts, being schedule decision influenced, and locality of cache. Its solution was to write a custom scheduling algorithm and insert it in the execution path of the model, as opposed to using a general optimization.
P4: Familiarity of issues (infrequently encountered issues)	The control of expert memory during inference time is an uncommon and novel issue, particularly in undergraduates. Current models presuppose that all professionals remain resident and do not pay attention to eviction or dynamic scheduling. These were unusual challenges that were encountered by the project in the design of a new middleware-based approach.
P5: Extent of applicable codes	The work was required to be based on several standards and best practices: the principles of design of deep learning, strong software engineering, limitations of the use of GPUs, and research ethics. Although there was no formal code applied, the project followed the usual practices in machine-learning research and reproducible experiments.
P6: Extent of stakeholder involvement and conflicting requirements	Various stakeholders did not fully agree with what they wanted: researchers desired to be able to run large models, developers were interested in the ability to deploy on relatively small hardware, institutions were interested in affordable compute. CacheFlow addresses these requirements by allowing large MoE models to be executed on the consumer-grade GPUs with no special infrastructure.
P7: Interdependence	Components of the system are well-connected. The scheduling of experts affects the memory usage, latency, cache behaviour, and accuracy directly. When one of the elements, e.g. expert capacity, is changed, the dynamics of the entire system change so one must be very careful in coordinating and designing the whole system.

Which attributes of Complex Engineering Activities (A) are involved?

Attribute of Complex Engineering Activities	Explanation (Justify that the attribute was present and you have addressed it sufficiently)
A1: Range of resources	The project also utilized multiple sources of resources: GPUs, PyTorch, quantization libraries, academic and industry grade papers. The combination of these various aspects was central to the development and testing of CacheFlow.
A2: Level of interactions	The team approached the supervisor and each other closely in order to design the system, overcome implementation challenges, and analyze findings. The scheduler had to be refined and findings checked through continuous discussion and coordination.

A3: Innovation	It presented a priority based expert scheduler that dynamically controls expert residency during inference. The re-engineering of the MoE architecture to add the scheduling logic represents an imaginative approach to using tools other than the regular ones.
A4: Consequences to society and the environment	The project reduces the computational cost and energy consumption by using large language models which can be efficiently executed by devices with limited access to resources. This facilitates greener deployments of AI especially where the infrastructure is minimal.
A5: Familiarity	The activities were new and not repeated: coming up with a custom expert scheduler, re-engineering a complicated neural architecture, and analyzing system behaviour in severe memory bounds. This required learnings far beyond typical coursework.

Comments: The project is a success in addressing a challenging engineering issue, which incorporates sophisticated analysis, competing constraints, innovation, and system-level design. CacheFlow demonstrates that Mixture-of-Experts models can be efficiently implemented on a low-memory usage device through the implementation of dynamic expert scheduling. It meets most of the demands of complicated engineering project and is a great accomplishment of an undergraduate level of work.

Approval of Supervisor	Approval of Examination Committee
Signature	Signature